3.5 Equations of Parallel and Perpendicular Lines

Name _____ Period ____ Date _____

Geometry 3.5 (30-minute plan) · Student sheet

I can…

- find a point that partitions a segment in a given ratio.
- use slopes to tell whether lines are parallel or perpendicular.
- write equations of parallel and perpendicular lines.

Vocabulary · fill in the missing word

Slope — the ratio of the _____ change (rise) to the horizontal change (run) between two points on a line

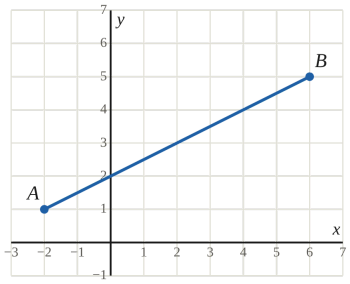
Directed line segment — a segment that has a direction from one _____ to the other

Part 1 · Notes and examples (fill in each note, then work the example)

Partitioning a segment: the point k of the way from $A(x_1, y_1)$ to $B(x_2, y_2)$ is $(x_1 + k(x_2 - x_1),$ _____$)$.

Example 1

Find the point P that is $\frac{3}{4}$ of the way from A(−2, 1) to B(6, 5).



Slopes of Parallel Lines Theorem: two nonvertical lines are parallel if and only if they have the _____ slope.

Example 2

Write an equation of the line through (3, 4) that is parallel to $y = 2x - 1$.

Slopes of Perpendicular Lines Theorem: two nonvertical lines are perpendicular if and only if the product of their slopes is _____.

Distance from a point to a line: write the perpendicular line through the point, find the _____ of the two lines, then use the Distance Formula.

Example 3

Write an equation of the line through $(2, 5)$ that is perpendicular to $y = 2x - 1$.

Part 2 · Partner practice

Partner practice

Write an equation of the line through $(4, -1)$ perpendicular to $y = (\frac{2}{3})x + 1$.